

Centralized

Requirement ID	Requirement Description	Design Elements	Test Cases
REQ-001	The system shall initialize in the Idle state.	Idle state in FloorController	Test that the system starts in Idle state
REQ-002	The system shall allocate an elevator upon receiving a request.	AllocateElevator method in FloorController	Test allocating an elevator when a request is received
REQ-003	The system shall open elevator doors when at a floor.	openDoors method in Elevator	Test opening doors when elevator reaches a floor
REQ-004	The system shall close elevator doors after a delay.	closeDoors method in Elevator	Test closing doors after delay
REQ-005	The system shall move the elevator up when requested.	moveUp method in Elevator	Test moving the elevator up
REQ-006	The system shall move the elevator down when requested.	moveDown method in Elevator	Test moving the elevator down
REQ-007	The system shall add a passenger to the elevator.	addPassenger method in Elevator	Test adding a passenger to the elevator
REQ-008	The system shall remove a passenger from the elevator.	removePassenger method in Elevator	Test removing a passenger from the elevator

Decentralized

Requirement ID	Requirement Description	Design Elements	Test Cases
REQ-001	The system shall initialize each elevator in the Idle state.	Idle state in Elevator class	Test that each elevator starts in Idle state
REQ-002	The system shall dispatch an elevator upon receiving a request.	DispatchElevator method in ElevatorSystem class	Test dispatching an elevator when a request is received
REQ-003	The system shall open elevator doors when at a floor.	openDoors method in Elevator class	Test opening doors when elevator reaches a floor
REQ-004	The system shall close elevator doors after a delay.	closeDoors method in Elevator class	Test closing doors after delay
REQ-005	The system shall move the elevator up when requested.	moveUp method in Elevator class	Test moving the elevator up
REQ-006	The system shall move the elevator down when requested.	moveDown method in Elevator class	Test moving the elevator down
REQ-007	The system shall add a passenger to the elevator.	addPassenger method in Elevator class	Test adding a passenger to the elevator
REQ-008	The system shall remove a passenger from the elevator.	removePassenger method in Elevator class	Test removing a passenger from the elevator